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Methods of training acoustic scene classification model and classifying acoustic scene and electronic device for performing the methods

Abstract

Disclosed are methods of training an acoustic scene classification model and classifying an acoustic scene and an electronic device for performing the methods. The training method of an acoustic scene classification model includes inputting training data labeled as an acoustic scene to the acoustic scene classification model that is repeatedly trained by using the training data and outputting a first result predicting the acoustic scene, updating the weight of the auxiliary model configured to induce training of the acoustic scene classification model, based on a weight of the acoustic scene classification model and a weight of an auxiliary model in a previous epoch, inputting the training data to the auxiliary model and outputting a second result, calculating a cost function, based on the first result, the second result, and labeling of acoustic data, and updating the weight of the acoustic scene classification model, based on the cost function.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims the priority benefit of Korean Patent Application No. 10-2021-0156679 filed on Nov. 15, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Field

(2) One or more example embodiments relate to methods of training an acoustic scene classification model and classifying an acoustic scene and an electronic device for performing the methods.

2. Description of Related Art

(3) Some techniques have been developed to improve the performance of an artificial neural network-based system. For example, a model-ensemble technique is used, in which the model-ensemble technique stores models obtained in a neural network training process, independently obtains an inference result of each of the stored models in an inference process, and combines the inference results together.

(4) In the model-ensemble technique, the complexity of a whole inference model increases as the number of models that are combined together increases. In an environment, such as a mobile environment, where the complexity of an inference model is limited, using the model-ensemble technique may not readily improve the performance of the inference model.

SUMMARY

(5) An aspect provides methods of training an acoustic scene classification model and classifying an acoustic scene and an electronic device for performing the methods, which may improve the performance of the acoustic scene classification model through an ensemble effect without increasing the complexity of an inference model in an environment where the complexity of the inference model is limited.

(6) Another aspect provides a training method of an acoustic scene classification model including inputting training data labeled as an acoustic scene to the acoustic scene classification model that is repeatedly trained by using the training data and outputting a first result predicting the acoustic scene, updating the weight of the auxiliary model configured to induce training of the acoustic scene classification model, based on a weight of the acoustic scene classification model and a weight of an auxiliary model in a previous epoch, inputting the training data to the auxiliary model and outputting a second result, calculating a cost function, based on the first result, the second result, and labeling of acoustic data, and updating the weight of the acoustic scene classification model, based on the cost function.

(7) The updating the weight of the auxiliary model may include, based on a reflection ratio of the weight of the acoustic scene classification model and the weight of the auxiliary model in a previous epoch, in which the reflection ratio is determined based on a weight coefficient, updating the weight of the auxiliary model.

(8) The updating the weight of the auxiliary model may include, as the weight coefficient increases, by increasing the reflection ratio of the weight of the acoustic scene classification model and decreasing the reflection ratio of the weight of the auxiliary model in a previous epoch, updating the weight of the auxiliary model.

(9) The updating the weight of the auxiliary model may include, as an epoch increases, increasing the weight coefficient up until the weight coefficient is less than or equal to a set maximum weight coefficient.

(10) The updating the weight of the auxiliary model may include, when an epoch is less than an epoch that is set to train the acoustic scene classification model alone, determining the weight coefficient such that the weight of the auxiliary model is updated based on the weight of the acoustic scene classification model.

- (11) The outputting the first result may include adding noise to the training data and inputting the noise-added training data to the acoustic scene classification model.
- (12) The calculating the cost function may include calculating a first cost function by using the first result and the labeling of the training data, calculating a second cost function by using the first result and the second result, and multiplying the second cost function by a cost function coefficient and adding the first cost function thereto.
- (13) The calculating the cost function may include, when an epoch is less than an epoch that is set to train the acoustic scene classification model alone, determining the cost function coefficient such that the cost function is calculated based on the first cost function.
- (14) Another aspect also provides a method of classifying an acoustic scene including identifying acoustic data including the acoustic scene, inputting the acoustic data to an acoustic scene classification model that is trained, and predicting the acoustic scene, in which the acoustic scene classification model is trained by using an auxiliary model configured to induce training of the acoustic scene classification model.
- (15) The acoustic scene classification model may be trained by inputting training data labeled as the acoustic scene to the acoustic scene classification model that is repeatedly trained by using the training data and outputting a first result predicting the acoustic scene, updating the auxiliary model, based on a weight of the acoustic scene classification model and a weight of the auxiliary model in a previous epoch, inputting the acoustic data to the auxiliary model and outputting a second result, calculating a cost function, based on the first result, the second result, and labeling of the acoustic data, and updating the weight of the acoustic scene classification model based on the cost function.
- (16) Another aspect also provides an electronic device including a processor, in which the processor is configured to identify acoustic data including an acoustic scene, input the acoustic data to an acoustic scene classification model that is trained, and predict the acoustic scene, in which the acoustic scene classification model is trained by using an auxiliary model configured to induce training of the acoustic scene classification model.
- (17) The acoustic scene classification model may be trained by inputting training data labeled as the acoustic scene to the acoustic scene classification model that is repeatedly trained by using the training data and outputting a first result predicting the acoustic scene, updating the auxiliary model, based on a weight of the acoustic scene classification model and a weight of the auxiliary model in a previous epoch, inputting the training data to the auxiliary model and outputting a second result, calculating a cost function, based on the first result, the second result, and labeling of the training data, and updating the weight of the acoustic scene classification model, based on the cost function.
- (18) The weight of the auxiliary model may be updated based on a reflection ratio of the weight of the acoustic scene classification model and the weight of the auxiliary model in a previous epoch, in which the reflection ratio is determined based on a weight coefficient.
- (19) The weight of the auxiliary model may be updated as the weight coefficient increases by increasing the reflection ratio of the weight of the acoustic scene classification model and decreasing the reflection ratio of the weight of the auxiliary model in a previous epoch.
- (20) The weight coefficient may increase as an epoch increases up until the weight coefficient is less than or equal to a set maximum weight coefficient.
- (21) The weight coefficient, when an epoch is less than an epoch that is set to train the acoustic scene classification model alone, may be determined such that the weight of the auxiliary model is updated based on the weight of the acoustic scene classification model.
- (22) The first result may be output by inputting the training data to which noise is added to the acoustic scene classification model.
- (23) The cost function may be calculated by calculating a first cost function by using the first result and the labeling of the training data, calculating a second cost function by using the first result and

the second result, multiplying the second cost function by a cost function coefficient, and adding the first cost function thereto.

(24) The cost function may be calculated based on the first cost function when an epoch is less than an epoch that is set to train the acoustic scene classification model alone.

(25) According to an aspect, the performance of a neural network-based acoustic scene classification model may be improved without increasing the complexity of an inference model.

(26) According to an aspect, the performance of an acoustic scene classification model may be improved in an environment, such as a mobile environment, where the complexity of a model is limited.

(27) According to an aspect, a method of training a neural network, such as an acoustic scene classification model, by using an auxiliary model may not transform an existing training model. The performance of a model may be improved by applying the method to the model that has been effectively operating.

(28) Additional aspects of example embodiments will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and/or other aspects, features, and advantages of the present disclosure will become apparent and more readily appreciated from the following description of example embodiments, taken in conjunction with the accompanying drawings of which:

(2) FIG. 1 is a diagram illustrating an operation of training an acoustic scene classification model of an electronic device, according to an example embodiment;

(3) FIG. 2 is a diagram illustrating an operation for a training method of the acoustic scene classification model according to an example embodiment;

(4) FIG. 3 is a diagram illustrating a whole training process of the acoustic scene classification model according to an example embodiment;

(5) FIG. 4 is a diagram illustrating an operation of classifying an acoustic scene using an electronic device according to an example embodiment; and

(6) FIG. 5 is a diagram illustrating a cost function coefficient and a learning rate, based on an epoch, according to an example embodiment.

DETAILED DESCRIPTION

(7) Hereinafter, example embodiments will be described in detail with reference to the accompanying drawings. However, various alterations and modifications may be made to the example embodiments. Here, the example embodiments are not construed as limited to the disclosure. The example embodiments should be understood to include all changes, equivalents, and replacements within the idea and the technical scope of the disclosure.

(8) The terminology used herein is for the purpose of describing particular example embodiments only and is not to be limiting of the example embodiments. The singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises/comprising” and/or “includes/including” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

(9) Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, such as those defined in commonly-

used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(10) When describing the example embodiments with reference to the accompanying drawings, like reference numerals refer to like constituent elements and a repeated description related thereto will be omitted. In the description of example embodiments, detailed description of well-known related structures or functions will be omitted when it is deemed that such description will cause ambiguous interpretation of the present disclosure.

(11) FIG. 1 is a diagram illustrating an operation of training an acoustic scene classification model **110** of an electronic device **100**, according to an example embodiment.

(12) Referring to FIG. 1, the electronic device **100** may train the acoustic scene classification model **110** by using an auxiliary model **120** for inducing the training of the acoustic scene classification model **110**. For example, the acoustic scene classification model **110** may be trained by using an ensemble effect caused by the auxiliary model **120** that is used.

(13) Referring to FIG. 1, the electronic device **100** may input training data **160** labeled as an acoustic scene to the acoustic scene classification model **110** and may output a first result predicting the acoustic scene.

(14) For example, the acoustic scene classification model **110** may be a neural network model, to which various known neural network models may be applied. For example, the neural network model may include a plurality of artificial neural network layers. An artificial neural network may include one of a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), and a bidirectional recurrent deep neural network (BRDNN), and a deep Q-network, but examples are not limited thereto. The neural network model may additionally or alternatively include a software structure other than a hardware structure.

(15) For example, the electronic device **100** may add noise **150** to the training data **160** and input the training data **160** to the acoustic scene classification model **110**. For example, the electronic device **100** may add the noise **150** to the training data **160** and train the acoustic scene classification model **110** such that each epoch in which the acoustic scene classification model **110** is trained may have variability.

(16) The electronic device **100** may update a weight of the auxiliary model **120**, based on a weight of the acoustic scene classification model **110** and a weight of the auxiliary model **120** in a previous epoch. By updating the weight of the auxiliary model **120**, based on the weight of the acoustic scene classification model **110** and the weight of the auxiliary model **120** in a previous epoch, the electronic device **100** may train the acoustic scene classification model **110**, based on a weighted average of the acoustic scene classification model **110** and the auxiliary model **120**.

(17) For example, the weight of the auxiliary model **120** may be updated based on the weight of the acoustic scene classification model **110**. For example, the auxiliary model **120** may be a model for inducing the training of the acoustic scene classification model **110**. By using the auxiliary model **120**, a model for training the acoustic scene classification model **110** may be labeled as an ensemble-inducing model.

(18) For example, the electronic device **100** may update the weight of the auxiliary model **120** to an exponential moving average (EMA) of weights of the acoustic scene classification model **110** or an exponential weighted moving average of the weights of the acoustic scene classification model **110**. The EMA and the like for updating the weight of the auxiliary model **120** are only an example, and examples of updating the weight of the auxiliary model **120**, based on the weight of the acoustic scene classification model **110**, are not limited to the foregoing examples. For example, the electronic device **100** may update the weight of the auxiliary model **120** by using a moving average of the weights of the acoustic scene classification model **110**.

(19) Equation 1 below may be an equation for updating the weight of the auxiliary model **120** by

the electronic device **100**. For example, referring to Equation 1, the electronic device **100** may update the weight of the auxiliary model **120**, based on a weight coefficient. For example, based on the weight coefficient, the electronic device **100** may determine a reflection ratio, to the weight of the auxiliary model **120** to be updated, of the weight of the acoustic scene classification model **110** and the weight of the auxiliary model **120** in a previous epoch.

$$\theta_{\text{sub.ema}} \leftarrow (1-\alpha)\theta_{\text{sub.ema}} + \alpha\theta \quad [\text{Equation 1}]$$

(20) In Equation 1, $\theta_{\text{sub.ema}}$ denotes the weight of the auxiliary model **120**, θ denotes the weight of the acoustic scene classification model **110**, and α denotes the weight coefficient. In Equation 1, $\theta_{\text{sub.ema}}$ on the right may be the weight of the auxiliary model **120** in a previous epoch and $\theta_{\text{sub.ema}}$ on the left may be the weight, to be updated, of the auxiliary model **120**.

(21) In Equation 1, $(1-\alpha)$ may be a reflection ratio of the weight of the auxiliary model **120** in a previous epoch, and the weight coefficient α may be a reflection ratio of the weight of the acoustic scene classification model **110**.

(22) Referring to Equation 1, the electronic device **100** may update the weight of the auxiliary model **120** by increasing the reflection ratio of the weight of the acoustic scene classification model **110** and decreasing the reflection ratio of the weight of the auxiliary model **120** in a previous epoch as the weight coefficient increases. In Equation 1, when the weight coefficient increases, the reflection ratio $(1-\alpha)$ of the weight of the auxiliary model **120** in a previous epoch may decrease and the reflection ratio α of the weight of the acoustic scene classification model **110** may increase.

(23) According to an example embodiment, the weight of the auxiliary model **120** may be updated to the weights of the acoustic scene classification model **110**, for example, to an EMA of the weights of the acoustic scene classification model **110** in a previous epoch, and the auxiliary model **120** may be referred to as an EMA model. A weight coefficient may be referred to as a moving average coefficient.

(24) In Equation 1, the weight coefficient α may be a constant for adjusting a valid length of a window function of the weight of the acoustic scene classification model **110**. For example, when the weight coefficient α increases, a weight of the acoustic scene classification model **110** in previous epochs may be less reflected in the weight of the auxiliary model **120** than a weight of the acoustic scene classification model **110** in a current epoch, and the weight of the acoustic scene classification model **110** in the current epoch may be more reflected in the weight of the auxiliary model **120** than the weight of the acoustic scene classification model **110** in the previous epochs. As the weight coefficient α increases, the weight of the auxiliary model **120** may more sensitively respond to the weight of the acoustic scene classification model **110** in the current epoch.

(25) In Equation 1, the weight of the auxiliary model **120** may be updated based on the weight of the acoustic scene classification model **110**, the weight of the auxiliary model **120** in a previous epoch, and the weight coefficient. In Equation 1, the weight of the auxiliary model **120** may be determined based on the weights of the acoustic scene classification model **110**.

(26) For example, in Equation 1, updating the weight of the auxiliary model **120**, based on the weight of the acoustic scene classification model **110** and the weight of the auxiliary model **120** in a previous epoch, may be understood to be the same as updating the weight of the auxiliary model **120**, based on an EMA of the weights of the acoustic scene classification model **110**.

(27) For example, the electronic device **100** may increase a weight coefficient, as an epoch increases, up until the weight coefficient is less than or equal to a set maximum weight coefficient.

$$\alpha = \min(1 - \frac{1}{n+1}, \alpha_{\text{sub.max}}) \quad [\text{Equation 2}]$$

(29) In Equation 2, α denotes a weight coefficient, $\alpha_{\text{sub.max}}$ denotes the set maximum weight coefficient, n denotes an epoch, and $\min(a, b)$ denotes a smaller value between a and b . In Equation 2, the weight coefficient α may be determined to be a smaller value between

$$(30) 1 - \frac{1}{n+1}$$

that is determined based on an epoch and the set maximum weight coefficient $\alpha_{\text{sub.max}}$.

(31) The electronic device **100** may input the training data **160** to the auxiliary model **120** and output a second result. For example, the second result that is output from the auxiliary model **120** may be practically in the same form as that of the first result, which is about a predicted acoustic scene.

(32) For example, the auxiliary model **120** may be a model for inducing the training of the acoustic scene classification model **110**, for example, a model for an ensemble-inducing training of the acoustic scene classification model **110**. For example, the auxiliary model **120** may include the same neural network structure as that of the acoustic scene classification model **110**.

(33) For example, the auxiliary model **120** may include the same neural network structure as that of the acoustic scene classification model **110** and may include the weight of the auxiliary model **120** that is determined based on the weight of the acoustic scene classification model **110**. The electronic device **100** may input the training data **160** to the auxiliary model **120** and may output the second result that is formally the same as the first result.

(34) The electronic device **100** may calculate a cost function by using the first result, the second result, and the labeling of the training data **160**.

(35) For example, the electronic device **100** may calculate a first cost function **130** by using the first result and the labeling of the training data **160**. For example, the electronic device **100** may calculate a second cost function **140** by using the first result and the second result. For example, the first cost function **130**, which is calculated by using a first output of the acoustic scene classification model **110** and the labeling of the training data **160**, may be referred to as a classification cost function. For example, the second cost function **140**, which is calculated by using the first output of the acoustic scene classification model **110** and a second output, may be referred to as a coherence cost function.

(36) For example, the first cost function **130** may compare a difference between the first result of the acoustic scene classification model **110** and labeling, that is, a correct answer, and the second cost function **140** may compare a difference between the first result of the acoustic scene classification model **110** and the second result of the auxiliary model **120**.

(37) For example, the electronic device **100** may calculate a cost function by using a sum of a weight of the first cost function **130** and a weight of the second cost function **140**. For example, the electronic device **100** may calculate the cost function by summing up a value of the first cost function **130** multiplied by a cost function coefficient and a value of the second cost function **140** multiplied by a cost function coefficient.

(38) For example, the electronic device **100** may calculate the cost function as Equation 3 below.
$$C(\theta) = C_{\text{sub.class}}(\theta) + \beta C_{\text{sub.consist}}(\theta) \quad [\text{Equation 3}]$$

(39) In Equation 3, $C(\theta)$ denotes the cost function, $C_{\text{sub.class}}(\theta)$ denotes the first cost function **130**, β denotes the cost function coefficient, and $C_{\text{sub.consist}}(\theta)$ denotes the second cost function **140**.

(40) The electronic device **100** may update the weight of the acoustic scene classification model **110** by using a cost function. For example, the electronic device **100** may update the weight of the acoustic scene classification model **110** to optimize the cost function.

(41) FIG. 2 is a diagram illustrating an operation for a training method of the acoustic scene classification model **110** according to an example embodiment.

(42) Referring to FIG. 2, in operation **210**, the electronic device **100** may input the training data **160** to the acoustic scene classification model **110** and output a first result. For example, the training data **160** may be acoustic data or an acoustic signal that is labeled as an acoustic scene. For example, the first result may be an acoustic scene predicted by using the training data **160** by the acoustic scene classification model **110** and may be data in a form corresponding to the labeling of the training data **160**.

(43) In operation **220**, the electronic device **100** may update a weight of the auxiliary model **120**, based on a weight of the acoustic scene classification model **110** and a weight of the auxiliary

model **120** in a previous epoch.

(44) For example, the weight of the auxiliary model **120** may be updated by using an EMA of weights of the acoustic scene classification model **110** in previous epochs.

(45) In operation **230**, the electronic device **100** may input the training data **160** to the auxiliary model **120** and output a second result.

(46) In operation **240**, the electronic device **100** may calculate a cost function by using the first result, the second result, and the labeling of the training data **160**. For example, the electronic device **100** may calculate the first cost function **130** by using a difference between the first result and the labeling of the training data **160** and may calculate the second cost function **140** by using a difference between the first result and the second result. The electronic device **100** may calculate a cost function by summing up a value of the first cost function **130** multiplied by a cost function coefficient and a value of the second cost function **140** multiplied by a cost function coefficient.

(47) In operation **250**, the electronic device **100** may update the weight of the acoustic scene classification model **110** by using a cost function. For example, the electronic device **100** may update a weight of the auxiliary model **120** in a next epoch by using the updated weight of the acoustic scene classification model **110**.

(48) FIG. **3** is a diagram illustrating a whole training process of the acoustic scene classification model **110** according to an example embodiment. Referring to FIG. **3**, the electronic device **100** may train the acoustic scene classification model **110**, without using the auxiliary model **120**, in an epoch that is less than a set epoch.

(49) For example, when the epoch is less than the set epoch, the electronic device **100** may train the acoustic scene classification model **110** without using the auxiliary model **120**. For example, when training the acoustic scene classification model **110** in an epoch that is less than the set epoch, the electronic device **100** may determine a weight coefficient and a cost function coefficient such that the acoustic scene classification model **110** may be trained alone. For example, the electronic device **100** may train the acoustic scene classification model **110** without operating an ensemble-inducing model in an epoch that is less than the set epoch and may train the acoustic scene classification model **110** by operating the ensemble-inducing model in an epoch that is greater than or equal to the set epoch.

(50) For example, training the acoustic scene classification model **110** alone in an epoch that is less than the set epoch may refer to updating a weight of the auxiliary model **120** to be the same as a weight of the acoustic scene classification model **110** and calculating a cost function by using the first cost function **130** calculated based on a difference between a first result and labeling of training data.

(51) Referring to FIG. **3**, in operation **310**, the electronic device **100** may determine a weight coefficient such that the weight of the auxiliary model **120** may be updated based on the weight of the acoustic scene classification model **110**.

(52) For example, the electronic device **100** may update the weight of the auxiliary model **120**, based on the weight of the acoustic scene classification model **110**, by setting the weight coefficient α in Equation 1 to 1.

(53) In operation **320**, the electronic device **100** may determine a cost function coefficient such that a cost function may be calculated based on the first cost function **130**.

(54) For example, the electronic device **100** may calculate the cost function, based on the first cost function **130**, by setting the cost function coefficient β in Equation 3 to 0.

(55) In operation **330**, the electronic device **100** may train the acoustic scene classification model **110**. For example, in operation **330**, the electronic device **100** may train the acoustic scene classification model **110** according to operations **210** to **250** illustrated in FIG. **2**. For example, in operation **330**, when the acoustic scene classification model **110** is trained once by using the training data **160**, an epoch may increase by 1.

(56) In operation **340**, the electronic device **100** may compare an epoch with a set epoch. For

example, the set epoch may refer to an interval where the acoustic scene classification model **110** is trained alone, for example, where an ensemble-inducing model does not operate.

(57) For example, in operation **340**, when the epoch is less than the set epoch, the electronic device **100** may perform operations **310** to **330**. For example, whenever the electronic device **100** repeats operations **310** to **330**, the epoch may increase by 1. The electronic device **100** may repeatedly perform operations **310** to **330** up until the epoch is greater than or equal to the set epoch.

(58) In operation **340**, when the epoch is greater than or equal to the set epoch, the electronic device **100** may determine a weight coefficient, based on the epoch and a set maximum weight coefficient. For example, the electronic device **100** may determine the weight coefficient based on Equation 2.

(59) In operation **360**, the electronic device **100** may set a cost function coefficient to a set value.

(60) In operation **370**, the electronic device **100** may train the acoustic scene classification model **110**. For example, in operation **370**, the electronic device **100** may train the acoustic scene classification model **110** according to operations **210** to **250** illustrated in FIG. 2. For example, in operation **370**, when the acoustic scene classification model **110** is trained once by using the training data **160**, an epoch may increase by 1.

(61) In operation **380**, the electronic device **100** may determine whether the training of the acoustic scene classification model **110** is completed. For example, the electronic device **100** may determine whether the training of the acoustic scene classification model **110** is completed by comparing an epoch with the set total number of training repetitions.

(62) For example, when the training is not completed in operation **380**, the electronic device **100** may train the acoustic scene classification model **110** according to operations **350** to **370**.

(63) For example, the electronic device **100** may set a learning rate differently in operations **330** and **370**. For example, the electronic device **100** may freely set the learning rate in operation **330** while setting the learning rate in operation **370** to low. For example, the electronic device **100** may set a learning rate of training of which an epoch is greater than or equal to the set epoch to a greater value than a learning rate of training of which an epoch is less than the set epoch.

(64) For example, the electronic device **100** may include a structure for training the acoustic scene classification model **110** by using the auxiliary model **120**, based on supervised learning. The electronic device **100** may improve the performance of a model by using an effect caused by the auxiliary model **120**, for example, by using an ensemble effect by an ensemble-inducing model. The electronic device **100** may effectively train the acoustic scene classification model **110** by training the acoustic scene classification model **110** alone in an epoch that is less than the set epoch as illustrated in FIG. 3.

(65) FIG. 4 is a diagram illustrating an operation of classifying an acoustic scene using an electronic device **400** according to an example embodiment.

(66) Referring to FIG. 4, the electronic device **400** may identify acoustic data **410** including an acoustic scene. The electronic device **400** may input the acoustic data **410** to the acoustic scene classification model **110** and output a predicted acoustic scene **430**.

(67) In FIG. 4, the acoustic scene classification model **110** may be trained by using the auxiliary model **120** for inducing the training of the acoustic scene classification model **110**. For example, the acoustic scene classification model **110** illustrated in FIG. 4 may be a neural network model trained according to the training method of the acoustic scene classification model **110** of the electronic device **100** illustrated in FIGS. 1, 2, and 3.

(68) FIG. 5 is a diagram illustrating a cost function coefficient and a learning rate, based on an epoch, according to an example embodiment. FIG. 5 is a diagram illustrating an example of updating a weight of the auxiliary model **120**, based on a weight of the acoustic scene classification model **110**, in an epoch that is less than a set epoch and calculating a cost function, based on the first cost function **130**, by the electronic device **100** described with reference to FIG. 3.

(69) FIG. 5 is an example where the set epoch is 300, in which the electronic device **100** may

calculate the cost function by setting the cost function coefficient β in Equation 3 to 0 in an epoch that is less than 300. The electronic device **100** may freely set a learning rate of the acoustic scene classification model **110** in an epoch that is less than 300.

(70) For example, an epoch interval where the cost function coefficient β is 0 may be referred to as an ensemble-inducing model non-operation area, and an epoch interval where the cost function coefficient β is not 0 may be referred to as an ensemble-inducing model operation area. For example, the electronic device **100** may set a learning rate, to be constant, in the ensemble-inducing model operation area where the cost function coefficient β is not 0.

(71) In another example different from the example illustrated in FIG. 5, the electronic device **100** may update the weight of the auxiliary model **120** by setting the weight coefficient α in Equation 1 to 1 in an epoch that is less than 300.

(72) For example, the electronic device **100** may train the acoustic scene classification model **110** by setting the cost function coefficient β to 0 or the weight coefficient α to 1 in an epoch that is less than the set epoch.

(73) For example, when the cost function coefficient β is 0, the cost function may be calculated by using the first cost function according to Equation 3. The electronic device **100** may update the weight of the acoustic scene classification model **110** to minimize the first cost function **130**. Therefore, when the cost function coefficient β is 0, the acoustic scene classification model **110** may be trained alone.

(74) For example, when the weight coefficient α is 1, the weight of the auxiliary model **120** may be updated to be the same as the weight of the acoustic scene classification model **110**. Since the weight of the auxiliary model **120** is the same as the weight of the acoustic scene classification model **110**, a first result may be the same as a second result and the second cost function **140** may be 0. Since the second cost function **140** is 0, the electronic device **100** may calculate the cost function according to the first cost function **130**, and when the weight coefficient α is 1, the acoustic scene classification model **110** may be trained alone.

(75) The method according to example embodiments may be written in a computer-executable program and may be implemented as various recording media such as magnetic storage media, optical reading media, or digital storage media.

(76) Various techniques described herein may be implemented in digital electronic circuitry, computer hardware, firmware, software, or combinations thereof. The implementations may be achieved as a computer program product, i.e., a computer program tangibly embodied in an information carrier, e.g., in a machine-readable storage device (for example, a computer-readable medium) or in a propagated signal, for processing by, or to control an operation of, a data processing apparatus, e.g., a programmable processor, a computer, or multiple computers. A computer program, such as the computer program(s) described above, may be written in any form of a programming language, including compiled or interpreted languages, and may be deployed in any form, including as a stand-alone program or as a module, a component, a subroutine, or other units suitable for use in a computing environment. A computer program may be deployed to be processed on one computer or multiple computers at one site or distributed across multiple sites and interconnected by a communication network.

(77) Processors suitable for processing of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random-access memory, or both. Elements of a computer may include at least one processor for executing instructions and one or more memory devices for storing instructions and data.

Generally, a computer also may include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. Examples of information carriers suitable for embodying computer program instructions and data include semiconductor memory devices, e.g., magnetic media such as hard

disks, floppy disks, and magnetic tape, optical media such as compact disk read only memory (CD-ROM) or digital video disks (DVDs), magneto-optical media such as floptical disks, read-only memory (ROM), random-access memory (RAM), flash memory, erasable programmable ROM (EPROM), or electrically erasable programmable ROM (EEPROM). The processor and the memory may be supplemented by, or incorporated in special purpose logic circuitry.

(78) In addition, non-transitory computer-readable media may be any available media that may be accessed by a computer and may include both computer storage media and transmission media.

(79) Although the present specification includes details of a plurality of specific example embodiments, the details should not be construed as limiting any invention or a scope that can be claimed, but rather should be construed as being descriptions of features that may be peculiar to specific example embodiments of specific inventions. Specific features described in the present specification in the context of individual example embodiments may be combined and implemented in a single example embodiment. On the contrary, various features described in the context of a single example embodiment may be implemented in a plurality of example embodiments individually or in any appropriate sub-combination. Furthermore, although features may operate in a specific combination and may be initially depicted as being claimed, one or more features of a claimed combination may be excluded from the combination in some cases, and the claimed combination may be changed into a sub-combination or a modification of the sub-combination.

(80) Likewise, although operations are depicted in a specific order in the drawings, it should not be understood that the operations must be performed in the depicted specific order or sequential order or all the shown operations must be performed in order to obtain a preferred result. In specific cases, multitasking and parallel processing may be advantageous. In addition, it should not be understood that the separation of various device components of the aforementioned example embodiments is required for all the example embodiments, and it should be understood that the aforementioned program components and apparatuses may be integrated into a single software product or packaged into multiple software products.

(81) The example embodiments disclosed in the present specification and the drawings are intended merely to present specific examples in order to aid in understanding of the present disclosure, but are not intended to limit the scope of the present disclosure. It will be apparent to those skilled in the art that various modifications based on the technical spirit of the present disclosure, as well as the disclosed example embodiments, can be made.

Claims

1. A training method of an acoustic scene classification model, the training method comprising: inputting training data, via a processor, labeled as an acoustic scene to the acoustic scene classification model that is repeatedly trained, via the processor, by using the training data and outputting a first result, via the processor, predicting the acoustic scene; based on a weight of the acoustic scene classification model in a current epoch and a weight of an auxiliary model in a previous epoch, updating the weight, via the processor, of the auxiliary model configured to induce training, via the processor, of the acoustic scene classification model; inputting the training data, via the processor, to the auxiliary model and outputting a second result via the processor; based on the first result, the second result, and the labeling of the training data, calculating a cost function via the processor; and based on the cost function, updating the weight, via the processor, of the acoustic scene classification model, wherein the updating the weight, via the processor, of the auxiliary model comprises; based on a reflection ratio of the weight of the acoustic scene classification model in a current epoch and the weight of the auxiliary model in a previous epoch, wherein the reflection ratio is determined based on a weight coefficient, updating the weight, via the processor, of the auxiliary model by the equation;

$\theta_{\text{sub.ema}} \leftarrow (1-\alpha)\theta_{\text{sub.ema}} + \alpha\theta$ wherein $\theta_{\text{sub.ema}}$ denotes the weight of the auxiliary model, θ denotes the weight of the acoustic scene classification model, and α denotes the weight coefficient, where, $\theta_{\text{sub.ema}}$ on the right of the equation is the weight of the auxiliary model in a previous epoch and $\theta_{\text{sub.ema}}$ on the left of the equation is the weight, to be updated, of the auxiliary model, wherein $(1-\alpha)$ is a reflection ratio of the weight of the auxiliary model in a previous epoch, and the weight coefficient α is a reflection ratio of the weight of the acoustic scene classification model, and wherein the weight coefficient α is defined by the equation;

$\alpha = \min(1-1/n+1, \alpha_{\text{sub.max}})$ wherein, $\alpha_{\text{sub.max}}$ denotes the set maximum weight coefficient, n denotes an epoch, and $\min(a, b)$ denotes a smaller value between a and b , and wherein the weight coefficient α is determined to be a smaller value between $1-1/n+1$ that is determined based on an epoch and the set maximum weight coefficient $\alpha_{\text{sub.max}}$.

2. The training method of claim 1, wherein the updating the weight, via the processor, of the auxiliary model comprises: as the weight coefficient increases, by increasing the reflection ratio, via the processor, of the weight of the acoustic scene classification model and decreasing the reflection ratio, via the processor, of the weight of the auxiliary model in a previous epoch, updating the weight, via the processor, of the auxiliary model.

3. The training method of claim 1, wherein the updating the weight, via the processor, of the auxiliary model comprises: as an epoch increases by 1, increasing the weight coefficient, via the processor, up until the weight coefficient is less than or equal to a set maximum weight coefficient.

4. The training method of claim 1, wherein the updating the weight, via the processor, of the auxiliary model comprises: when the number of epochs is less than a set number of epochs that is set to train, via the processor, the acoustic scene classification model alone, determining the weight coefficient, via the processor, such that the weight of the auxiliary model is updated based on the weight of the acoustic scene classification model.

5. The training method of claim 1, wherein the outputting the first result, via the processor, comprises: adding noise to the training data, via the processor, and inputting the noise-added training data, via the processor, to the acoustic scene classification model.

6. The training method of claim 1, wherein the calculating the cost function, via the processor, comprises: calculating a first cost function, via the processor, by using the first result and the labeling of the acoustic scene; calculating a second cost function, via the processor, by using the first result and the second result; and multiplying the second cost function, via the processor, by a cost function coefficient and adding the first cost function, via the processor, to the second cost function multiplied by the cost function coefficient via the processor, by the equation:

$C(\theta) = C_{\text{sub.class}}(\theta) + \beta C_{\text{sub.consist}}(\theta)$ wherein $C(\theta)$ denotes the cost function, $C_{\text{sub.class}}(\theta)$ denotes the first cost function, β denotes the cost function coefficient, and $C_{\text{sub.consist}}(\theta)$ denotes the second cost function.

7. The training method of claim 6, wherein the calculating the cost function, via the processor, comprises: when the number of epochs is less than a set number of epochs that is set to train, via the processor, the acoustic scene classification model alone, determining the cost function coefficient, via the processor, such that the cost function is calculated based on the first cost function via the processor.

8. A method of classifying an acoustic scene, the method comprising: identifying acoustic data, via a processor, comprising the acoustic scene; and inputting the acoustic data to an acoustic scene classification model, via the processor, that is trained and predicting the acoustic scene via the processor, wherein the acoustic scene classification model is trained, via the processor, by using an auxiliary model configured to induce training of the acoustic scene classification model, wherein the acoustic scene classification model is trained, via the processor, by inputting training data, via the processor, labeled as the acoustic scene to the acoustic scene classification model that is repeatedly trained, via the processor, by using the training data and outputting a first result, via the processor, predicting the acoustic scene, updating a weight of the auxiliary model, via the

processor, based on a weight of the acoustic scene classification model in a current epoch and a weight of the auxiliary model in a previous epoch, inputting the training data, via the processor, to the auxiliary model and outputting a second result via the processor, calculating a cost function, via the processor, based on the first result, the second result, and labeling of the training data, and updating the weight of the acoustic scene classification model via the processor, based on the cost function, and wherein the cost function is calculated, via the processor, by the equation;
 $C(\theta) = C_{\text{sub.class}}(\theta) + \beta C_{\text{sub.consist}}(\theta)$ wherein $C(\theta)$ denotes the cost function, $C_{\text{sub.class}}(\theta)$ denotes a first cost function, β denotes a cost function coefficient, and $C_{\text{sub.consist}}(\theta)$ denotes a second cost function.

9. An electronic device comprising: a processor, wherein the processor is configured to identify acoustic data comprising an acoustic scene, input the acoustic data to an acoustic scene classification model that is trained, and predict the acoustic scene, wherein the acoustic scene classification model is trained, via the processor, by using an auxiliary model configured to induce training of the acoustic scene classification model, wherein the acoustic scene classification model is trained, via the processor, by inputting training data, via the processor, labeled as the acoustic scene to the acoustic scene classification model that is repeatedly trained by using the training data and outputting a first result predicting the acoustic scene, updating a weight of the auxiliary model, via the processor, based on a weight of the acoustic scene classification model in a current epoch and a weight of the auxiliary model in a previous epoch, inputting the training data, via the processor, to the auxiliary model and outputting a second result via the processor, calculating a cost function, via the processor, based on the first result, the second result and labeling of the training data, and updating the weight of the acoustic scene classification model via the processor, based on the cost function, and wherein the cost function is calculated, via the processor, by the equation:
 $C(\theta) = C_{\text{sub.class}}(\theta) + \beta C_{\text{sub.consist}}(\theta)$ wherein $C(\theta)$ denotes the cost function, $C_{\text{sub.class}}(\theta)$ denotes a first cost function, β denotes a cost function coefficient, and $C_{\text{sub.consist}}(\theta)$ denotes a second cost function.

10. The electronic device of claim 9, wherein the weight of the auxiliary model is updated, via the processor, based on a reflection ratio of the weight of the acoustic scene classification model in a current epoch and the weight of the auxiliary model in a previous epoch, wherein the reflection ratio is determined, via the processor, based on a weight coefficient.

11. The electronic device of claim 10, wherein the weight of the auxiliary model is updated, via the processor, as the weight coefficient increases by increasing the reflection ratio of the weight of the acoustic scene classification model and decreasing the reflection ratio of the weight of the auxiliary model in a previous epoch.

12. The electronic device of claim 10, wherein the weight coefficient increases as an epoch increases by 1 until the weight coefficient is less than or equal to a set maximum weight coefficient.

13. The electronic device of claim 10, wherein the weight coefficient, when the number of epochs is less than a set number of epochs that is set to train the acoustic scene classification model alone, via the processor, is determined such that the weight of the auxiliary model is updated based on the weight of the acoustic scene classification model.

14. The electronic device of claim 9, wherein the first result is output, via the processor, by inputting the training data to which noise is added to the acoustic scene classification model.

15. The electronic device of claim 9, wherein the cost function is calculated by calculating the first cost function, via the processor, by using the first result and the labeling of the acoustic scene, calculating the second cost function, via the processor, by using the first result and the second result, multiplying the second cost function, via the processor, by the cost function coefficient, and adding the first cost function, via the processor, to the second cost function multiplied by the cost function coefficient.

16. The electronic device of claim 15, wherein the cost function is calculated, via the processor,

based on the first cost function when the number of epochs is less than a set number of epochs that is set to train the acoustic scene classification model alone.
